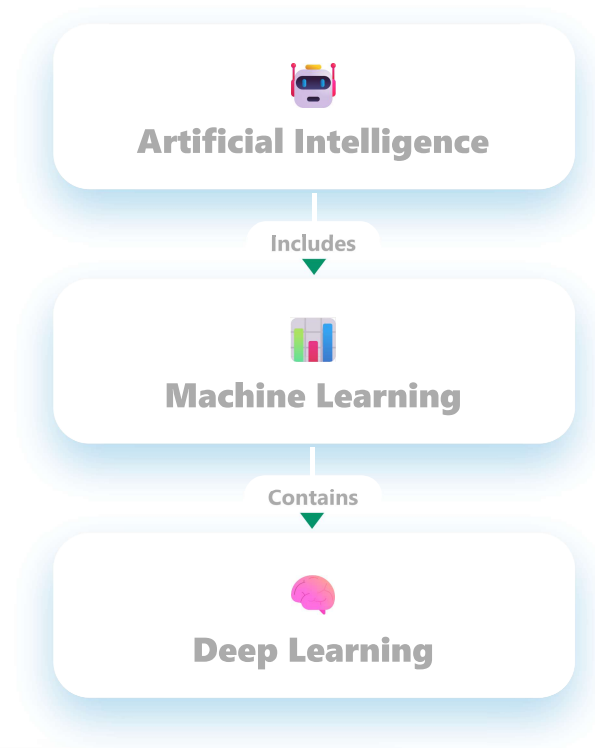


CONCEPT MAP · GENERATIVE AI

A Comparative Concept Map under Artificial Intelligence

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Machine Learning

DEFINITION

Algorithms learn patterns from data.

REQUIRES

Feature Engineering: Manual · **Training Data:** Moderate

COMPUTATION

Low to Medium

TRAINING TIME

Faster

EXAMPLES

Spam Detection, Recommendation Systems

 ALGORITHMS

Decision Tree Random Forest SVM KNN

APPLICATIONS

Fraud Detection Email Filtering Sales Prediction

VSNeeds fewer data ⇌
Needs large dataManual ⇌ Automatic
feature extractionLower ⇌ Higher
computational costTraditional algorithms
⇌ Neural networks ⚡**Deep Learning****DEFINITION**

Uses multi-layer neural networks to learn automatically.

REQUIRES**Feature Engineering:** Automatic · **Training Data:** Large Datasets**COMPUTATION**

Very High (GPU/TPU)

TRAINING TIME

Longer

EXAMPLES

Face Recognition, ChatGPT, Self-driving Cars

**MODELS**

CNN RNN Transformer

APPLICATIONS

Image Recognition Speech Recognition Language Translation Generative AI

 Both branches share the same source of data — but differ in how deeply they learn from it.**CONCLUSION**

"Deep Learning is a specialized subset of Machine Learning that automatically learns complex patterns from large amounts of data."

REAL-WORLD EXAMPLES**Machine Learning**

- Netflix Recommendation
- Email Spam Filter

**Deep Learning**

- ChatGPT
- Google Translate
- Tesla Autopilot

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